

Stratigraphic-Aware Lithology Classification

and Uncertainty-Aware Grade Prediction

Julián D. Alfonso-Gómez

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FORCE 2020 (Norwegian well log benchmark) · PyTorch · XGBoost baseline

The Problem with Per-Sample Classifiers

Standard lithology models classify each depth sample **independently**:

$$\hat{y}_t = f(\mathbf{x}_t), \quad t = 1, \dots, L$$

What they miss: a shale layer sitting *above* a sandstone is a stratigraphic relationship — geological signal encoded in depth ordering, not in any single sample.

Our approach. Replace the pointwise classifier with a Transformer encoder that attends across the entire depth sequence:

$$\hat{\mathbf{Y}} = \text{TransEnc}(\mathbf{X}, \mathbf{d}) \in \mathbb{R}^{L \times C}$$

where $\mathbf{d} \in \mathbb{R}^L$ are **actual measured depths in metres** and $C = 11$ lithology classes.

FORCE 2020 — Norwegian Well Log Benchmark

Scale. 11 labeled wells · 121,424 depth samples at 0.1524 m intervals.

Input log curves ($n_f = 14$):

Curve	Signal	Curve	Signal
GR	Gamma Ray	DTC	Compressional Δt
RDEP	Deep Resistivity	DTS	Shear Δt
RMED	Medium Resistivity	PEF	Photoelectric
RHOB	Bulk Density	CALI	Caliper
NPHI	Neutron Porosity	DRHO	Density Correction

Plus 4 derived features: V_p/V_s , neutron-density crossplot, $R_{\text{dep}}/R_{\text{med}}$, per-well normalised GR.

Class Distribution — Severe Imbalance

Class	Count	%
Shale	69,305	57.1
Sandstone	14,794	12.2
Shale with sand	10,507	8.7
Limestone	8,721	7.2
Anhydrite	6,498	5.4
Marl	5,266	4.3
Limestone w/ clay	2,905	2.4
Coal	2,366	1.9
Halite	597	0.5
Chalk	269	0.2
Tuff	196	0.2

Shale alone is 57% of samples. A trivial classifier predicting only shale gets 57% accuracy but Macro F1 ≈ 0.09 — this is why we use focal loss.

Stage 1 — Transformer Architecture

Step	Operation
Input	$\mathbf{X} \in \mathbb{R}^{B \times L \times n_f}$ — well log curves
Projection	Linear $\mathbb{R}^{n_f} \rightarrow \mathbb{R}^{d_{\text{model}}}$, $d_{\text{model}} = 128$
Positional encoding	Depth-aware sinusoidal, added to projection
Encoder	$N = 4$ Transformer layers, $H = 4$ heads, $d_{\text{ff}} = 256$
Head	Linear $\mathbb{R}^{d_{\text{model}}} \rightarrow \mathbb{R}^C$
Output	$\hat{\mathbf{Y}} \in \mathbb{R}^{B \times L \times 11}$ — per-interval logits

Sliding window: $L = 128$ depth samples (≈ 19.5 m), stride = 64, batch $B = 64$.

Depth-Aware Positional Encoding

Standard Transformers encode *sequence position* t . We encode **actual measured depth** d_t **in metres**, so a 10 m interval at 500 m differs from one at 2,000 m:

$$\text{PE}(d_t, 2i) = \sin(d_t \cdot \omega_i), \quad \text{PE}(d_t, 2i + 1) = \cos(d_t \cdot \omega_i)$$

$$\omega_i = \exp\left(-\frac{2i}{d_{\text{model}}} \ln D_{\text{max}}\right), \quad D_{\text{max}} = 6000 \text{ m}$$

The encoded token at position t :

$$\tilde{\mathbf{x}}_t = \underbrace{W_{\text{in}} \mathbf{x}_t}_{\text{log projection}} + \underbrace{\text{PE}(d_t)}_{\text{depth info}} \in \mathbb{R}^{d_{\text{model}}}$$

Absolute depth matters geologically: evaporites at > 3 km, shallow clastics at < 1 km. Irregular sampling is handled naturally.

Scaled Dot-Product Attention

Given $\tilde{\mathbf{X}} \in \mathbb{R}^{L \times d_{\text{model}}}$, project to queries, keys, values:

$$\mathbf{Q} = \tilde{\mathbf{X}}W^Q, \quad \mathbf{K} = \tilde{\mathbf{X}}W^K, \quad \mathbf{V} = \tilde{\mathbf{X}}W^V$$

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \in \mathbb{R}^{L \times d_k}$$

The attention matrix $\mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d_k}) \in [0, 1]^{L \times L}$ encodes **how much depth interval t attends to interval s** . Each row sums to 1.

Scaling by $1/\sqrt{d_k}$ keeps dot product variance at unity, preventing softmax saturation and vanishing gradients.

Multi-Head Attention

H heads run in parallel, each with its own projections $W_h^Q, W_h^K, W_h^V \in \mathbb{R}^{d_{\text{model}} \times d_k}$ where $d_k = d_{\text{model}}/H$:

$$\text{head}_h = \text{Attn}(\tilde{\mathbf{X}}W_h^Q, \tilde{\mathbf{X}}W_h^K, \tilde{\mathbf{X}}W_h^V)$$

$$\text{MHA}(\tilde{\mathbf{X}}) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W^O, \quad H = 4, d_k = 32$$

Each encoder layer wraps MHA in a residual block:

$$\mathbf{Z} = \text{LayerNorm}(\tilde{\mathbf{X}} + \text{MHA}(\tilde{\mathbf{X}}))$$

$$\mathbf{Z}' = \text{LayerNorm}(\mathbf{Z} + \text{FFN}(\mathbf{Z}))$$

Different heads specialise in different stratigraphic relationships simultaneously — one head may capture resistivity patterns, another density-porosity relationships.

Focal Loss for Class Imbalance

Standard cross-entropy for a sample with true class c :

$$\mathcal{L}_{\text{CE}} = -\log \hat{p}_c$$

Problem: shale is 57% of samples. A model that always predicts shale gets $\mathcal{L}_{\text{CE}} \approx 0.56$ — low enough that gradient descent stalls before learning minority classes.

Focal loss adds a modulating factor $(1 - \hat{p}_c)^\gamma$:

$$\mathcal{L}_{\text{FL}} = -(1 - \hat{p}_c)^\gamma \log \hat{p}_c, \quad \gamma = 2$$

Easy correct predictions ($\hat{p}_c = 0.95$): factor = 0.0025, near zero contribution. Hard or wrong predictions ($\hat{p}_c = 0.1$): factor = 0.81, dominates the gradient.

Automatic curriculum learning — the model focuses capacity on what it doesn't yet know.

Spatial Cross-Validation

Random row-level splits are **catastrophically optimistic** for well log data:

- **Autocorrelation:** rock at 1500.00 m is nearly identical to rock at 1499.85 m. If one is in train and the other in test, the model memorises rather than generalises.
- **Spatial correlation:** nearby wells share basin geology. Splitting nearby wells across folds inflates apparent performance.

Our protocol. Cluster wells by surface coordinates (X_{LOC} , Y_{LOC}) via k -means. Hold out entire clusters as folds. The model has *never seen* the held-out geographic region during training.

Harder evaluation, but it answers the question that matters in deployment: can the model predict lithology in geology it has never seen?

Stage 1 Results — Single Held-Out Well

Best checkpoint, 50 epochs, validation well 31/2-21 S:

Model	Macro F1	Sandstone	Shale	Shale/Sand	Limestone
Transformer	0.4149	0.721	0.820	0.355	0.594
XGBoost	0.5157	0.713	0.870	0.190	0.615

Transformer wins on transitional facies (Shale with sand, +0.165 F1) — context-dependent classes where stratigraphic sequence matters most. This is the architectural thesis validated.

XGBoost leads on ultra-rare classes (Tuff, Coal) where the Transformer overfits on a handful of examples.

Stage 1 Results — 5-Fold Spatial CV

15 epochs per fold, geographically isolated held-out wells:

Model	Macro F1 (mean)	Macro F1 (std)
Transformer	0.2660	± 0.061
XGBoost	0.2522	± 0.109

Half the variance across folds. The Transformer is more consistent across unseen geological environments — exactly what matters when deploying to new mine sites.

Calibration: **ECE** = 0.080. The model's confidence estimates are reasonably trustworthy, which makes them usable as soft conditioning signals for downstream prediction.

Stage 2 — Cross-Attention Grade Predictor

Branch	Inputs and operations
Depth branch	Assay values + Stage 1 lithology probabilities → Transformer encoder → $\mathbf{Z}_d \in \mathbb{R}^{L \times d_{\text{model}}}$
Spatial branch	Collar (x, y, z) + surface geophysics → MLP → $\mathbf{z}_s \in \mathbb{R}^{d_{\text{model}}}$
Fusion	Cross-attention: $\mathbf{z}_s$ queries $\mathbf{Z}_d$
Output	Regression head → grade $\hat{g}$ + uncertainty

The spatial context queries the depth sequence — the model asks *which stratigraphic intervals are most predictive of grade at this surface location?*

Cross-Attention Fusion — Mathematics

The spatial context vector queries the depth sequence:

$$\begin{aligned}\mathbf{Q}_s &= \mathbf{z}_s W_s^Q \in \mathbb{R}^{1 \times d_{\text{model}}} \\ \mathbf{K}_d &= \mathbf{Z}_d W_d^K, \quad \mathbf{V}_d = \mathbf{Z}_d W_d^V \in \mathbb{R}^{L \times d_{\text{model}}} \\ \alpha_\ell &= \frac{\exp(\mathbf{Q}_s \mathbf{k}_\ell^T / \sqrt{d_k})}{\sum_{\ell'} \exp(\mathbf{Q}_s \mathbf{k}_{\ell'}^T / \sqrt{d_k})}, \quad \mathbf{f} = \sum_{\ell=1}^L \alpha_\ell \mathbf{v}_\ell \in \mathbb{R}^{d_{\text{model}}} \\ \hat{g} &= \text{RegressionHead}(\mathbf{f})\end{aligned}$$

Geological interpretation. $\alpha \in [0, 1]^L$ is a depth-importance profile — the weights are directly interpretable by geologists. Cross-attention learns that mineralised alteration halos at specific depths matter more than simple concatenation could express.

Uncertainty via Deep Ensembles

Grade prediction in mineral exploration is **high-stakes**: uncertainty quantification is as important as the point estimate.

Train $M = 5$ independent models with different random seeds:

$$\bar{g} = \frac{1}{M} \sum_{m=1}^M \hat{g}^{(m)}, \quad \sigma_{\text{ens}}^2 = \frac{1}{M} \sum_{m=1}^M \left(\hat{g}^{(m)} - \bar{g} \right)^2$$

Benchmarked against ordinary kriging variance:

$$\sigma_{\text{OK}}^2(\mathbf{x}_0) = C(0) - \boldsymbol{\lambda}^T \mathbf{c}_0$$

Lakshminarayanan et al. (2017) show ensembles outperform MC Dropout in calibration, especially in low-data regimes typical of mineral exploration.

Summary

Contributions

- Depth-sequence Transformer with self-attention over *actual measured depths*, not sequence indices.
- +0.165 **F1** on transitional facies (Shale with sand) vs XGBoost.
- $\sigma_{\text{fold}} = 0.061$ vs XGBoost's 0.109 — more spatially stable.
- Stage 2 cross-attention gives interpretable depth-importance weights for grade prediction.
- ECE = 0.080 — calibrated probabilities suitable as soft conditioning signals.

Next steps

- Train Stage 2 on GSWA Western Australian drill database (Au, Cu, Ni, Zn).
- Pre-train on all 21 FORCE wells via masked-depth modelling (semi-supervised).
- Active learning loop using σ_{ens}^2 to recommend next drill location.

Thank you

github.com/cyber-pocho/stat_aware_pred

References

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